THESIS REVISION GUIDE
Combined Checklist

Reassessed against the local thesis source and supporting files on 2 November 2026. Open items are unchecked.

3.2.2 Bregman and β-divergences

☐ **Repair the general β-divergence setup.** Reconcile the real-valued domain and regularity assumptions with the power generator for general real $\beta > 0$. Check the domain separately from the valid quadratic special case $\beta = 1$. [chapter lines 165–183]

☐ **Align the Bregman symmetry statement with its source.** The statement currently omits the Legendre-type generator assumption of the cited Tsuchida–Nielsen theorem. Establish the needed assumptions or revise the statement so its scope is justified. [chapter lines 159–163]

3.2 Numerical approximation

☐ **Verify the interval count used for the reported main experiments.** The current quadrature defaults to $K = 1376$, matching the numerical-approximation study. The archived validation reports older Fourier-grid settings for Figures 3.1–3.4. Verify provenance of the currently referenced figures and report the settings actually used; do not assume the current default generated all saved results. [chapter lines 213–238]

Final check

☐ **Compile the finished thesis.** After the substantive revisions, check the sections, figures, and cross-references.

Resolved: the plug-in notation and citation details, including the AMISE exponent and Appendix reference, have been corrected. The earlier consistency and oracle items have also been addressed. Reassessment covers the prior checklist and newly revised plug-in passage. Earlier citation cleanup and other sections marked 02.11 remain outside its scope. Source line numbers may shift.

REASSESSED | BANDWIDTH GRID AND THEORY

3.5.2 Propositions 8 and 9

☐ **Check the propositions for the geometric grid.** The experiments use $r = 0$, but Proposition 8 assumes $0 < r \leq 1$ and Proposition 9 uses the same setup. Check whether both proofs extend to $r = 0$; if they do, state the extension. Otherwise, qualify the connection to the experiments. [chapter lines 466, 586–619]

Resolved: the grid definition takes the limit of finite differences, the appendix covers $r = 0$, and the spacing lemma now uses a signed difference and indices $N \geq 2$. The original $h_1, \ldots, h_N$ convention is retained.

REASSESSED | NUMERICAL EXPERIMENTS

3.4 Presentation of the numerical experiments

☐ **Define the experimental setup and comparisons.** Specify the sampling distribution using $X_i \sim N(0,1)$, the kernel, grid and endpoints, sample sizes and repetitions, and how the selected bandwidth is obtained. Distinguish the population MISE oracle, the sample ISE minimizer, and the energy minimizer; state whether each is continuous or grid-restricted. [chapter lines 417–427, 478–508]

☐ **Make the fixed-sample figure descriptions match the plots.** The first profile is described as $i \mapsto v_i$ and as $\lambda = 1$, whereas its plot uses bandwidth on the horizontal axis and shows four λ values. The grid-fineness caption lists only $\lambda = 1$ while the plot shows four curves. Reconcile text, legends, reference lines and captions with the saved figures. [chapter lines 429–450]

☐ **Qualify the concentration interpretation.** Explain precisely what the entropy-penalized optimization establishes for individual normalized rows, and distinguish that result from the observed behavior of the Perron vector as λ changes. [chapter lines 349–354, 419–431]

☐ **Reconcile the joint-limit claim with the figure and theory.** The text and caption say convergence for all $\alpha > 1/2$, including 2 and 3, but the figure labels those regimes "mesh-scale" and "collapse"; the later proposition assumes $\lambda_N/N^2 \to 0$. Check each curve and qualify the claim rather than treating finite-grid behavior as a proof of a limit. [chapter lines 466–475, 586–593]

☐ **Correct the energy-functional notation.** The joint-limit paragraph calls the displayed $h^2\|\partial_h \hat{f}_h\|_2^2/2$ functional $\hat{Q}_{200,2}$, while the later grid-weighted definition identifies this power with $r = 0$ and the plot uses $\hat{Q}_0$. Align the notation, the geometric grid and the continuous reference. [chapter lines 466, 593]

☐ **Make the repeated-sampling comparison self-contained.** Explain the two λ settings, M, n, N, the five-grid-point bins and all compared estimators, then describe each panel rather than making one unqualified discrepancy claim. Give the parent figure and its right-hand subfigure distinct labels so references resolve unambiguously. [chapter lines 484–508]

☐ **Resolve the remaining editorial notes.** Turn the red question about jumps and the inline notes on N^2, N^3, alternative grids and repeated sampling into supported interpretation, clearly identified future work, or a decision to omit. Dedicated fixed-sample and repeated-sampling notebooks exist; assess the remaining scope before keeping their notebook TODOs. [chapter lines 423–425, 454–482]

Evidence: bandwidth-selection chapter, appendix, referenced figure PDFs, KDE notebooks and archived numerical validation. Source line numbers may shift.